# Jiten Dhandha

Kavli Fellow - IPMU, University of Tokyo

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## Research interests

My research focuses on the study of the early Universe, spanning the **Dark Ages, Cosmic Dawn, and the Epoch of Reionization** (redshift $z \sim 6 - 50$). My expertise lies in the theoretical modelling of the **cosmic 21-cm signal** of neutral hydrogen, and of **stellar and galactic astrophysics**; this includes numerical simulations of present-day ($z \sim 0$) Population I star-formation in molecular clouds, as well as analytic and semi-numerical simulations of high-redshift ($z \gtrsim 6$) Population II and Population III stars. I have extensively used **machine learning and Bayesian inference techniques** to accelerate simulations and perform parameter inference, combining data from multi-wavelength facilities (e.g. from JWST and 21-cm experiments).

## Employment

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| 2026 - *present* | **Kavli Postdoctoral Fellow**, Kavli Institute for Physics and Mathematics of the Universe (Kavli IPMU), University of Tokyo. |

## Education

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| 2022 - 2026 | **PhD in Astronomy**, Institute of Astronomy, University of Cambridge. Funded by Boustany Astronomy Scholarship & Isaac Newton Studentship at Pembroke College. Supervised by Prof. Anastasia Fialkov and Dr. Eloy de Lera Acedo. **Thesis**: Constraining the Early Universe with 21-cm and Multi-Wavelength Inference in the JWST and SKA Era. |
| 2018 - 2022 | **MPhys. Physics with Astrophysics** (First Class), University of Manchester. Project involved simulating turbulent molecular clouds in ISM and studying filament and star formation. Performed with Zoe Faes and supervised by Dr. Rowan Smith. |

## Internships

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| Jul. 2024 - Aug. 2024 | **Summer volunteer internship**, Boustany Foundation, Monaco. Partnering with Open Cultural Center, a humanitarian NGO focused on providing teaching, advice and extracurricular activities to asylum-seekers and refugees in the Nea Kavala camp in Greece. |
| Jun. 2021 - Aug. 2021 | **Summer research project**, University of Manchester. Modelling the cosmological 21-cm signal in `Recfast++` and `CosmoTherm` to study their synergy with CMB spectral distortions. Supervised by Prof. Jens Chluba. |
| Jun. 2020 - Sep. 2020 | **Summer research project**, University of Manchester. Testing and debugging `LOFAR-VLBI` calibration/imaging pipeline for gravitational lenses. Supervised by Dr. Neal Jackson. |
| Jul. 2019 - Sep. 2019 | **Summer intern programme**, British Petroleum / University of Manchester. Simulating mitigation techniques for sulphate reducing bacteria responsible for fouling crude oil. Supervised by Dr. Thomas Waigh. |

## Grants and awards

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| April 2024 | **DiRAC Resource Allocation Committee 16th Call**, awarded 4.15M CPUh (worth £41,500) on DiRAC's COSMA-8 supercomputer. |
| July 2022 | **Tessella Prize for Software** (£125), for outstanding work implementing software in Mphys project. |
| March 2022 | **Boustany Scholarship for Astronomy** ($\sim$ £100,530), for PhD at University of Cambridge. |
| April 2019 | **BP Achievement Award** (£1000), for best essay on petrophysical logging tools. |
| December 2018 | **Physics Success Scholarship** (£2000), for academic excellence in physics and maths. |

# Services

### Conference, Workshop and Meeting organization

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| March 2026 - *present* | 21-cm Meeting Series, co-organiser of bi-weekly meetings at KICC, University of Cambridge, focused on the cosmological 21-cm signal and related topics. |
| May 2026 | Galaxies and the 21-cm Frontier: Unveiling the Epoch of Reionization (Kavli Science Focus), member of Organising Committee and session chair, KICC, University of Cambridge. |
| July 2025 | Radio cosmology and science with the 21-cm signal (one day workshop), member of Organising Committee and session chair, KICC, University of Cambridge. |
| February 2024 | Science with the 21-cm line (Kavli Science Focus), member of Organising Committee and session chair, KICC, University of Cambridge. |

### Other committees

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| Feb. 2025 - Mar. 2025 | **International Womens Day Committee** member, Institute of Astronomy, UoC. |
| May 2023 - Oct. 2025 | **EDI Inclusion and Fairness** subgroup member, Institute of Astronomy, UoC. |
| Jul. 2023 - Jul. 2024 | **Graduate Parlour**, Ethnic Minorities officer, Pembroke College, UoC. |
| Oct. 2022 - Oct. 2025 | **Postgraduate Forum** representative, Institute of Astronomy, UoC. |
| Oct. 2022 - Apr. 2023 | **Pembroke Papers** committee memeber, Pembroke College, UoC. |

# Teaching responsibilities

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| Oct. 2025 - Sep. 2026 | **Co-Supervision: Seowon Jung** (Part III and summer student) with Prof. Anastasia Fialkov and Prof. Alexander Belyaev. Work on probing the nature of dark matter with 21-cm cosmology. |
| Oct. 2025 - Jul. 2026 | **Co-Supervision: Saughn Sekhon** (Part III student) with Prof. Anastasia Fialkov and Dr. Peter Sims. Work on machine learning accelerated Bayesian inference of early Universe star-formation. |
| Oct. 2024 - Jul. 2025 | **Co-Supervision: Jacques Valkenberg** (MPhil. student) with Prof. Anastasia Fialkov and Prof. Sandro Tacchella. Work on studying the impact of metal enrichment and Population II stellar IMF on the 21-cm signal of neutral hydrogen. |
| Oct. 2024 - Jul. 2025 | **Co-Supervision: Kyle Wong** (Part III student) with Prof. Anastasia Fialkov. Work on studying the impact of varying cosmology and matter power spectrum on the 21-cm signal of neutral hydrogen. |
| Oct. 2023 - Jul. 2024 | **Co-Supervision: Jamie Incley** (Part III student) with Prof. Anastasia Fialkov. Work on comparison of Epoch of Reionization in simulation codes `21cmSPACE` and `C2-Ray`. |
| Feb. 2023 - Mar. 2023 | **Demonstration of Part IA Scientific Computing** for 22 hours, University of Cambridge. |

# Talks

### Invited talks

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| April 2026 | *Constraining Galaxies in the Early Universe using 21-cm and Multi-Wavelength Synergies* (virtual), State of the Universe (SOTU) seminar, DTP, TIFR-Mumbai. |
| November 2025 | *Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints*, Weekly seminar, Institut d'Astrophysique Spatiale, Université Paris-Saclay. |
| September 2025 | *An overview of 21-cm cosmology and how we can constrain the discovery space of the 21-cm signal* (virtual), 8th Global 21-cm Workshop, California Institute of Technology. |
| October 2022 | *Can accreting primordial black holes explain the excess radio background?* (virtual), Weekly webinar, PDAT Laboratory, K. N. Toosi University of Technology. |

### Conference and Workshop talks

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| November 2026 | *Rotating Pop III stars and their impact on the high-z Universe*, Simulations GEARS, Kavli Institute for Physics and Mathematics of the Universe, University of Tokyo. |
| March 2026 | *Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints*, Cosmology in the Alps 2026, Les Diablerets, Switzerland. |
| June 2025 | *Exploiting synergies between JWST and cosmic 21-cm observations to uncover star formation in the early Universe*, SKAO General Science Meeting 2025, Görlitz, Germany. |

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| October 2024 | *Constraining star-formation efficiency in the early Universe using JWST and the cosmic 21-cm signal*, Introduction to KICC, Kavli Institute for Cosmology, University of Cambridge. |
| September 2024 | *Constraining star-formation efficiency in the early Universe using JWST and the cosmic 21-cm signal*, 7th Global 21-cm Workshop, Raman Research Institute. |
| May 2024 | *Synergies between 21-cm experiments and JWST observations*, Reionization in Relic Radiation (R3), Institut d'Astrophysique Spatiale, Université Paris-Saclay. |
| February 2024 | *Synergies between 21-cm experiments and JWST observations*, One day workshop: Science with the 21-cm line, Kavli Institute for Cosmology, University of Cambridge. |
| September 2023 | *FIlamEntary STructure Analysis (fiesta)*, AREPO-ISM workshop, University of Manchester. |

### Meeting talks

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| October 2026 | *What do we know about the cosmic 21-cm signal so far, and how can we improve upon it?*, Astro seminar, Kavli Institute for Physics and Mathematics of the Universe, University of Tokyo. |
| April 2026 | *Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints*, NenuFAR Cosmic Dawn plenary, Kavli Institute for Cosmology, University of Cambridge. |
| October 2025 | *An overview of 21-cm cosmology and how we can constrain the discovery space of the 21-cm signal*, mini-workshop on 21-cm signal (headed by Prof. Naoki Yoshida), Department of Physics, University of Tokyo. |
| September 2025 | *What do we know about the 21-cm signal so far?*, Gravitational Wave Astrophysics Group group meeting (headed by Prof. Michela Mapelli), Centre for Astronomy of Heidelberg University. |
| September 2023 | *Bringing 21-cm simulations to the JWST era*, 2nd REACH Annual Meeting, University of Malta. |

### Outreach talks

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| October 2022 | *Like beads on a string... Where do massive stars in our Universe come from? A brief look into studying our cosmos*, Pembroke Papers, Pembroke College, University of Cambridge. |

## Workshops attended

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| September 2025 | IWR School on Machine Learning for Fundamental Physics, Faculty of Mathematics and Computer Science, Heidelberg University. |

## Software development

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| `eor_limits` | Author and maintainer of the new beta-release of the web-app: eorlimits.streamlit.app, which allows users to compare upper limits on the 21-cm power spectrum from different experiments. |
| `21cmSimulators` | Author and maintainer: Comprehensive public repository of popular 21-cm signal simulation codes. |
| `21cmExperiments` | Author and maintainer: Comprehensive public repository of past and ongoing 21-cm experiments. |
| `CFit` | Author and maintainer: Smart curve fitting tool using method of least squares in Python. |
| `fiesta` | Author and maintainer: Toolkit for analyzing filament networks and density field meshes. |

## In the media

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| August 2021 | Most detailed-ever images of galaxies revealed using LOFAR. Press release for LOFAR observations from ASTRON. |
| August 2021 | Astronomers develop novel way to 'see' first stars through fog of early Universe. Press release for LOFAR observations from BBC. |

## Extracurricular activities

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| Sep. 2021 - Jul. 2022 | **Student Representative** representing astronomy/astrophysics, UoM. |
| Jul. 2020 - Jul. 2022 | **Touch Rugby Society**, Inclusion officer and COVID-19 safety officer, UoM. |
| Sep. 2019 - Jun. 2020 | **Peer-Assisted Study Session** leader, Peer Support Scheme,UoM. |

Nov. 2016 - *present* | **English Wikipedia**, volunteer editor.

# Publications

I have **3 first author** publications and **15 contributing author** publications. I have also contributed by peer-reviewing **1** paper.

### First Author

December 2025 | **J. Dhandha**, A. Fialkov, T. Gessey-Jones, H. T. J. Bevins, S. Tacchella, S. Pochinda, E. de Lera Acedo, S. Singh, R. Barkana, *Narrowing the discovery space of the cosmological 21-cm signal using multi-wavelength constraints*, Monthly Notices of the Royal Astronomical Society, 544, 1608

September 2025 | **J. Dhandha**, A. Fialkov, T. Gessey-Jones, H. T. J. Bevins, S. Tacchella, S. Pochinda, E. de Lera Acedo, S. Singh, R. Barkana, *Exploiting synergies between JWST and cosmic 21-cm observations to uncover star formation in the early Universe*, Monthly Notices of the Royal Astronomical Society, 542, 2292

April 2024 | **J. Dhandha**, Z. Faes, R. J. Smith, *Decaying turbulence in molecular clouds: how does it affect filament networks and star formation?*, Monthly Notices of the Royal Astronomical Society, 529, 4699

### Contributing Author

October 2026 | S. Dasgupta, B. Liu, A. Fialkov, F. Deng, **J. Dhandha**, R. Barkana, *Impact of Stochastic Pop III X-ray Binaries on the Cosmological 21-cm Signal*, Monthly Notices of the Royal Astronomical Society,

June 2026 | S. A. K. Leeney et al., *Scalable Bayesian data curation for next-generation radio experiments*, arXiv e-prints, arXiv:2606.16525

April 2026 | J. Wasserman, E. Zackrisson, **J. Dhandha**, A. Fialkov, L. Noble, S. Majumdar, *Ultraviolet photon production rates of the first stars: Impact on the He II λ 1640 Å emission line from primordial star clusters and the 21-cm signal from cosmic dawn*, Monthly Notices of the Royal Astronomical Society, 547, stag386

March 2026 | A. K. Dash et al., *Optimisation of calibration sources for global 21-cm experiments: the REACH case*, arXiv e-prints, arXiv:2604.00105

February 2026 | Z. Abdurashidova et al., *First Results from HERA Phase II*, The Astrophysical Journal, 998, 33

— 2026 | G. V. C. Allen et al., *Circuit Modeling for In Situ 21 cm Radiometer Calibration*, Journal of Astronomical Instrumentation, 15, 2650002-70

— 2026 | A. Chakraborty, T. R. Choudhury, K. K. Datta, P. Dayal, **J. Dhandha**, S. Gagnon-Hartman, S. K. Giri, A. Gorce, C. Heneka, A. Hutter, B. Maity, S. Majumdar, A. Mesinger, K. Moriwaki, C. S. Murmu, Y. Qin, S. Yoshiura, E. Zackrisson, *Synergies for the Epoch of Reionization and Cosmic Dawn*, Advancing Astrophysics with the SKA I, 137

October 2025 | S. Munshi, F. G. Mertens, J. K. Chege, L. V. E. Koopmans, A. R. Offringa, B. Semelin, R. Barkana, **J. Dhandha**, A. Fialkov, R. Mériot, S. Sikder, A. Bracco, S. A. Brackenhoff, E. Ceccotti, R. Ghara, S. Ghosh, I. Hothi, M. Mevius, P. Ocvirk, A. K. Shaw, S. Yatawatta, P. Zarka, *Improved upper limits on the 21-cm signal power spectrum at $z = 17.0$ and $z = 20.3$ from an optimal field observed with NenuFAR*, Monthly Notices of the Royal Astronomical Society, 542, 2785

August 2025 | B. Liu, D. Kessler, T. Gessey-Jones, **J. Dhandha**, A. Fialkov, Y. Sibony, G. Meynet, V. Bromm, R. Barkana, *Effects of chemically homogeneous evolution of the first stars on the 21-cm signal and reionization*, Monthly Notices of the Royal Astronomical Society, 541, 3113

March 2025 | O. S. D. O'Hara, Q. Gueuning, E. de Lera Acedo, F. Dulwich, J. Cumner, D. Anstey, A. Brown, A. Fialkov, **J. Dhandha**, A. Faulkner, Y. Liu, *Uncovering the effects of array mutual coupling in 21-cm experiments with the SKA-Low radio telescope*, Monthly Notices of the Royal Astronomical Society, 538, 31

February 2025 | S. Pochinda, **J. Dhandha**, A. Fialkov, E. de Lera Acedo, *Cosmological super-resolution of the 21-cm signal*, Accepted at NeurIPS 2024 (Machine Learning and the Physical Sciences Workshop), arXiv:2502.00852

September 2024 | O. S. D. O'Hara, F. Dulwich, E. de Lera Acedo, **J. Dhandha**, T. Gessey-Jones, D. Anstey, A. Fialkov, *Understanding spectral artefacts in SKA-Low 21-cm cosmology experiments: the impact of cable reflections*, Monthly Notices of the Royal Astronomical Society, 533, 2876

May 2024 | A. Fialkov, T. Gessey-Jones, **J. Dhandha**, *Cosmic mysteries and the hydrogen 21-cm line: bridging the gap with lunar observations*, Philosophical Transactions of the Royal Society of London Series A, 382, 20230068

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| December 2022 | S. K. Acharya, **J. Dhandha**, J. Chluba, *Can accreting primordial black holes explain the excess radio background?*, Monthly Notices of the Royal Astronomical Society, 517, 2454 |
| February 2022 | S. Badole, D. Venkattu, N. Jackson, S. Wallace, **J. Dhandha**, P. Hartley, C. Riddell-Rovira, A. Townsend, L. K. Morabito, J. P. McKean, *High-resolution imaging with the International LOFAR Telescope: Observations of the gravitational lenses MG 0751+2716 and CLASS B1600+434*, Astronomy and Astrophysics, 658, A7 |

## White Papers / Reports / Book Chapters

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| June 2026 | A. Mesinger et al., *Synergies for the Epoch of Reionization and Cosmic Dawn*, Advancing Astrophysics with the SKA II (AASKAII), 2026 |
| November 2025 | E. de Lera Acedo et al., *21 cm Cosmology from Space: A UK Path via CosmoCube*, UK Space Frontiers 2035 |
| November 2025 | J. Chluba et al., *Opening a New Window for Cosmology with CMB Spectral Distortions*, UK Space Frontiers 2035 |